Indian Institute of Technology Ropar
Department of Mathematics
MA634: Financial Risk Management
1st Semester of Academic Year 2026-27
Assignment

Deadline: November 13, 2026 (11:59 pm)

Instructions:

- Students may submit the assignment in groups of up to three members. Include the names and entry numbers of all group members in the report; only one submission per group is needed.
- Students, please be advised that any form of cheating or plagiarism, including copying from peers or the internet, will result in 0 marks. The assignments will be screened through **Plagiarism Checker by Grammarly**.
- Please email your solutions to **puneet.pasricha@iitrpr.ac.in** by November 13, 2026 at 23h59. The subject of the e-mail should be **"MA634 Assignment Your Entry Number"**. Your solutions should be neatly presented. Attach Python codes named in the format **"Assignment_EntryNumber_Codes.py"** and a PDF file named **"Assignment_EntryNumber_Solutions.pdf"**. The Python codes should execute without any issues and should be well-commented.
- There will be a viva after the submission deadline is over. The dates will be announced later.

Objective

(i) Construct Global Minimum Variance (GMV), Mean–Variance Efficient (tangency), Equal-Weight (EW), and an "Active" portfolio (based on alpha significance), (ii) run a rolling 6-month formation / 3-month holding backtest from 2009–2022, (iii) evaluate and compare the out-of-sample performance of these portfolios against the Nifty 50 benchmark.

Data You will find two csv files on the google classroom.

- *Stocks_data.csv*: Daily closing prices for *50* Nifty 50 constituents from **2009-01-01** to **2022-12-31**. The last column contains the closing value of Nifty 50. If a stock has missing/partially observations in the analysis period, you can remove that stock.
- *market_Factor_risk_Free.csv*: Daily factors: columns `Date, MF, RF`, where `MF` is the market excess return (in percentage) and `RF` is the daily risk-free rate (decimal).

Problems Unless stated otherwise, use *daily* simple returns (not log returns) in raw series and compound appropriately when aggregating.

1. **(Common dates)** Match trading dates across the stock, index, and factor files and **retain only the intersection** of dates for all subsequent analysis.Compute daily simple returns from consecutive available trading prices before merging them with the factor data; avoid using future information.
2. **(Portfolio construction from formation window)** For any formation window, let $R_t \in \mathbb{R}^N$ be the vector of stock returns on day t for the available N stocks in that window. Estimate the sample mean $\hat{\mu} \in \mathbb{R}^N$ and covariance $\hat{\Sigma} \in \mathbb{R}^{N\times N}$. Create the following four portfolios. For all four stock portfolios, use long-only, fully invested weights: $w_i \geq 0$ and $\sum_i w_i = 1$. Determine all weights using formation-window information only.

   a) **Global Minimum Variance (GMV) portfolio.** Solve the constrained optimization problem

$$\min_{w\in\mathbb{R}^N} w^\top \hat{\Sigma} w \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0.$$

b) **Mean–Variance (Tangency) portfolio.** Let $\bar{r}_f$ be the arithmetic mean of the daily risk-free rates RF_t in the formation window. Numerically maximize the daily estimated Sharpe ratio:

$$\max_{w \in \mathbb{R}^N} \frac{w^\top \hat{\mu} - \bar{r}_f}{\sqrt{w^\top \hat{\Sigma} w}} \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0.$$

State the numerical solver used.

c) **Equal-Weighted (EW) portfolio.** Set $w_i = 1/N$ for every stock in the chosen universe.

d) **Active portfolio via alpha significance (formation window).** For each stock i, run the CAPM regression on *excess* returns (with all series in decimals) using formation-window data:

$$(R_{i,t} - RF_t) = \alpha_i + \beta_i \, \mathrm{MF}_t + \varepsilon_{i,t}.$$

Use a two-sided test of $H_0 : \alpha_i = 0$ at the 5 % significance level. Define $\mathcal{A} = \{i : \hat{\alpha}_i > 0 \text{ and the two-sided } p\text{-value} < 0{,}05\}$. If $\mathcal{A} \neq \varnothing$, form a *long-only equal-weight active portfolio* with

$$w_i^{\text{Active}} = \begin{cases} 1/|\mathcal{A}|, & i \in \mathcal{A}, \\ 0, & i \notin \mathcal{A}. \end{cases}$$

If no stock qualifies, invest 100 % in the Nifty 50 index for that holding period (the Active return then equals the benchmark return). Do not take short positions in significantly negative-alpha stocks. Briefly acknowledge the multiple-testing limitation when testing many stocks.

3. Using the weights from the formation window, compute the realized portfolio return over the **holding period** (no rebalancing during the holding period). In particular, if the holding period has L trading days, the correct buy-and-hold cumulative simple return is

$$R_{p,\text{hold}} = \sum_{i=1}^{N} w_i^{(0)} \left[\prod_{d=1}^{L} (1 + R_{i,d}) - 1 \right],$$

where $w^{(0)}$ denotes weights at the end of formation. Do not treat the portfolio as daily constant-weight rebalanced. For the Nifty 50 benchmark (and Active when using the fallback), compound its daily index returns.

4. **(Rolling windows)** Begin with formation **Jan–Jun 2009** and holding **Jul–Sep 2009**. Shift the formation and holding **forward by 3 months** (rolling window): formation **Apr–Sep 2009**, holding **Oct–Dec 2009**; then **Jul–Dec 2009** / **Jan–Mar 2010**; continue until the last holding period **Oct–Dec 2022**. For each window, compute the 3-month realized returns for the **five** portfolios: [GMV, MV, EW, Active, NIFTY50].

5. **(Return matrix)** Stack the window results into an $n \times 5$ matrix $\mathbf{R}$, where n is the number of rolling windows. You should obtain $n = 54$.

6. **(Performance reporting)**

   a) Plot **cumulative returns** of all five portfolios across windows by compounding the 3-month holding returns sequentially.

   b) Create a **performance table** reporting, for each portfolio: mean return, standard deviation, **Sharpe ratio** (use an annualized definition and state your risk-free convention), and **Information ratio** relative to Nifty 50. Clearly state annualization choices (e.g., 252 trading days/year; 3 months $\approx$ 63 trading days) and how you convert window returns to annualized figures (e.g., via log-returns).

   c) Briefly **discuss** the results (risk–return trade-offs, performance relative to Nifty 50, etc.).

Assumptions & Notes

- Use fully invested, long-only stock portfolios ($\sum_{i=1}^{N} w_i = 1$, $w_i \geq 0$); the Nifty 50 index is allowed only as the stated Active fallback and as the benchmark.
- Be explicit about return definitions (simple vs. log), compounding conventions, and annualization constants. Convert all returns to decimals; keep unit consistency. Use daily simple returns throughout portfolio calculations. The formation/holding window design must not be changed.

Deliverables

1. A concise PDF report including: methodology summary, portfolio construction and optimization details (including the selected weights), cumulative return plot, performance table, and key takeaways.

2. Working code. The code must implement the rolling portfolio construction and buy-and-hold returns without look-ahead bias.

3. An $(n \times 5)$ CSV (or table in the appendix) of rolling 3-month holding returns with columns: GMV, MV, EW, Active, NIFTY50.